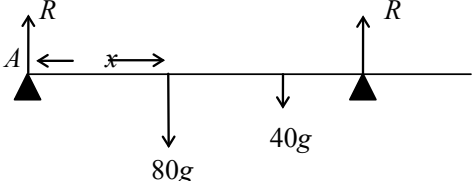
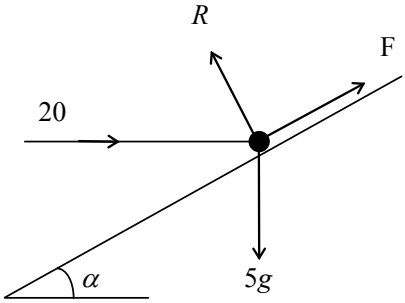


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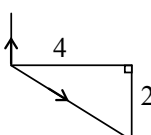
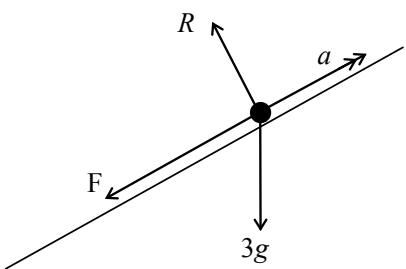
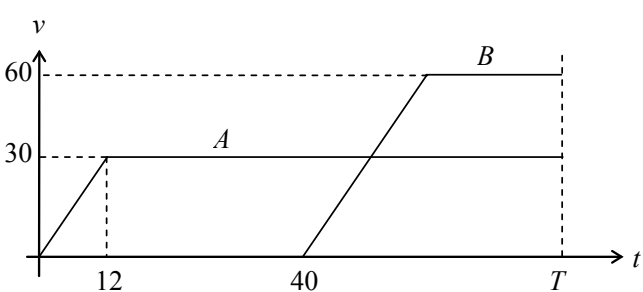
PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
1.	(a)  $R(\uparrow): 2R = 80g + 40g$ $R = 60g$ or 588 N (b) $M(A): 80g \times x + 40g \times 2 = 60g \times 3$ $\Rightarrow x = 1\frac{1}{4}$ m	M1 A1 (2) M1 A2 ft (-1 eeo) (4) (6 marks)
2.	(a) $I = 0.12 \times 3 = 0.36$, Ns (b) $0.12 \times 3 = 0.12 \times 1.2 + 0.08v$ $\Rightarrow v = 2.7$ m s⁻¹ (c) $I = 0.12 \times (3 - 1.2)$ or 0.08×2.7 $= 0.216$ Ns	B1, B1 (2) M1 A1 A1 (3) M1 A1 (2) (7 marks)
3.	(a) “$v^2 = u^2 + 2as$”: $v^2 = 4^2 + 2 \times g \times 5$ $v \approx 10.7$ m s⁻¹ (accept 11 m s⁻¹) (b) “$v = u + at$”: $-10.7 = 4 - gt$ $t = \frac{14.7}{g} = 1.5$ s (c) Air resistance; ‘spin’; height of diver; hit board again; horizontal component of velocity (any two)	M1 A1 A1 (3) M1 A1 ft A1 (3) B1 B1 (2) (8 marks)
4.	 $R(\backslash): R = 5g \cos \alpha + 20 \sin \alpha$ $R(\swarrow): F + 20 \cos \alpha = 5g \sin \alpha$ Using $\cos \alpha = \frac{4}{5}$ or $\sin \alpha = \frac{3}{5}$ $[\Rightarrow R = 51.2$ N; $F = 13.4$ N] Using $F = \mu R$ Solving: $\mu = 0.262$ (accept 0.26)	M1 A1 M1 A1 B1 M1 M1 A1 (8) (8 marks)

(ft = follow through mark; -1eeoo = minus one mark for each error or omission)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
5.	(a) “$v = u + at$”: $\mathbf{v} = (-2 + 2t)\mathbf{i} + (7 - 3t)\mathbf{j}$ $\mathbf{v}$ parallel to $\mathbf{i} \Rightarrow 7 - 3t = 0 \Rightarrow t = 2\frac{1}{3} \text{ s}$ (b) $t = 3, \mathbf{v} = 4\mathbf{i} - 2\mathbf{j}$ $\mathbf{v} = \sqrt{20} \approx 4.47 \text{ m s}^{-1}$ (c)  Angle = $(\arctan \frac{2}{4}), +90^\circ = 116.6^\circ$ (accept 117°) [or $180^\circ - (\arctan \frac{4}{2})$]	M1 A1 M1 A1 (4) M1 M1 A1 (3) M1, M1 A1 (3) [M1 M1 A1] (10 marks)
6.	(a)  $R(\backslash): R = 3g \cos 30^\circ (= 25.46 \text{ N})$ $F = 0.4R \approx 10.2 \text{ N}$ (accept 10 N) (b) $R(\nearrow): -F + 3g \sin 30^\circ = 3a$ $\Rightarrow a \approx 8.3 \text{ m s}^{-2}$ “$v^2 = u^2 + 2as$”: $6^2 = 2 \times a \times s$ $\Rightarrow s \approx 2.17 \text{ m}$ (accept 2.2 m)	M1 A1 M1 A1 (4) M1 A2 (-1 eeo) M1 A1 M1 A1 (7) (11 marks)
7.	(a)  Shape for A Shape for B with parallel slope Figures (b) Distance moved by A = $\frac{1}{2} \times 12 \times 30 + 30(T - 12)$ B accelerates for 24 s Distance moved by B = $\frac{1}{2} \times 24 \times 60 + 60(T - 64)$ $\frac{1}{2} \times 12 \times 30 + 30(T - 12) = \frac{1}{2} \times 24 \times 60 + 60(T - 64)$ $\Rightarrow T = 98 \text{ s}$	B1 B1 B1 (3) B1, M1 A1 B1 B1, M1 A1 M1 A1 (9) (12 marks)

(ft = follow through mark; -1 eeo = minus one mark for each error or omission)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
8. (a)	Car + truck: $2000a = 2400 - 600 - 400$ $a = 0.7 \text{ m s}^{-2}$	M1 A1 A1 (3)
(b)	Car only: $T - 400 = 800 \times 0.7$ [or truck only: $2400 - T - 600 = 1200 \times 0.7$] $T = 960 \text{ N}$	M1 A1 ft A1 (3)
(c)	New acceleration of truck a' given by $1200 a' = 2400 - 600$ $a' = 2400 - 600 = 1.5 \text{ m s}^{-1}$ Time to reach $28 \text{ m s}^{-1} = \frac{28 - 20}{1.5} = 5.33 \text{ s}$ Time to reach 28 m s^{-1} if rope had not broken = $\frac{28 - 20}{0.7} = 11.43 \text{ s}$ Difference = $6.1 \text{ s} \approx 6 \text{ s} (*)$	M1 A1 M1 A1 M1 A1 A1 (7) (13 marks)

(ft = follow through mark; (*) indicates final line is given on the paper)